

Luis A. Rodriguez

Technical Architect – Embedded & Web Systems

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Summary

Senior Technical Architect with 20+ years of experience in full-stack software development, spanning embedded systems, mobile gaming, and large-scale web applications used by millions. Expert in bridging low-level hardware emulation and proprietary protocols with modern web technologies like WebAssembly, TypeScript, and Angular. Proven technical leader and empathetic mentor, skilled in defining architectural requirements, driving code quality, and leading cross-functional teams to deliver robust, user-centered solutions.

Core Technical Skills

Languages	TypeScript, JavaScript, C, C++, Angular, Java, z80 Assembly, CSS, HTML
Architectural Core	WebAssembly, Emscripten, Cross-Compilation, Hardware Emulation, JNI, Lua Glue, USB Communication
Web & Frameworks	Angular, Spring Boot, WebSerial, WebUSB, RESTful APIs, Single-Sign-On (SSO)
DevOps & Tools	Gulp, Jenkins, CMake, MSBuild, Git, Node.js, Mocha, Selenium WebDriver
Leadership	Technical Leadership, Scrum Master, Agile/Scrum, Test-Driven Development (TDD)

Professional Experience

Texas Instruments -- Software Architect -- (2010 – 2025, Dallas, TX)

Progressive tenure specializing in bridging low-level hardware emulation and proprietary communication protocols with modern web applications and client product lines. Recognized with an appointment to the first rung of TI's Technical Ladder.

- **Architectural Leadership & Web-Native Integration:** Led development and served as Scrum Master for calculator web applications. Architected and implemented a unified file-transfer API using WebSerial for both physical and emulated calculator devices.
- **Pioneering Web Technologies (WebAssembly):** Designed and implemented the core glue layer to target embedded C code on the web, successfully transpiling to WebAssembly via Emscripten for the TI-84 Plus CE Online Calculator.
- **DevOps & Build System Optimization:** Developed and maintained Gulp.js build scripts for complex toolchains (CMake, MSBuild). Optimized the workflow into parallel Jenkins jobs, resulting in a 70% reduction in overall build time.
- **Massive User Adoption:** Led the rapid-response effort to develop and launch the TI-84 Plus CE Chrome App, which has since been used by over 3.5 million users.
- **Systems Engineering:** Established cross-platform USB communication architecture using a JNI abstraction layer to deliver platform-native libraries (Windows/macOS) for the TI Connect CE utility.
- **Accessibility & Quality:** Implemented low-level code in z80 Assembly and JavaScript to integrate calculator emulators with screen readers, ensuring compliance and accessibility for vision-impaired users.

Digital Chocolate -- Game Engineering Manager -- (2009 – 2010)

Managed the team responsible for porting mobile games for US carriers' devices, focusing on cost reduction and technical excellence.

- Coordinated project planning to reduce production costs and improve revenue. Led and mentored the engineering team, providing tutorials and code deep-dives.
- Created a client module for Qualcomm's Application Value Billing API (C/C++) to enable in-app purchases and promote code reuse across future game titles.

LemonQuest / Gameloft -- Senior Programmer Roles -- (2005 – 2009)

Focused on mobile game development, porting, and systems programming for the iPhone OS and resource-limited devices.

- Led the team that developed the first successful iPhone game prototype, leading to the studio being assigned more high-profile projects.
- Ported games from DirectX to iPhone's OpenGL ES, optimizing the user experience by translating desktop inputs for touch and accelerometer use.

Education & Professional Development**Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)**

- Completed coursework towards: Master's in Electronics and Telecommunications

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

- Bachelor's in Electronics Systems Engineering